

TOPS TECH Task 8 (CS)

Task-8 Check Live system, open ports, Services

- 1. Use the netstat command on your Windows or Linux system to list all currently open ports and their associated services, then take a screenshot of the output.**

Ans:

List Currently Open Ports and Associated Services

Objective:

- To identify the currently open ports and their associated services running on the system using the Netstat command.

Command:

→ netstat -ano

- The netstat -ano command displays all active network connections, listening ports, protocol information, connection states, and the Process ID (PID) associated with each connection. It is commonly used for network monitoring, troubleshooting, and security analysis.

Observations:

- The command output showed multiple active and listening ports on the system.
- These ports are used by various Windows services and applications for network communication.

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The following open ports were identified:

<u>Port Number</u>	<u>Protocol</u>	<u>State</u>	<u>Description</u>
135	TCP	LISTENING	Microsoft RPC (Remote Procedure Call) Service
445	TCP	LISTENING	SMB (Server Message Block) File Sharing Service
5040	TCP	LISTENING	Windows Push Notification Service
7680	TCP	LISTENING	Windows Update Delivery Optimization
139	TCP	LISTENING	NetBIOS Session Service
2020	TCP	LISTENING	Local Application Service
2021	TCP	LISTENING	Local Application Service
2022	TCP	LISTENING	Local Application Service

- ⇒ Several connections were found in the ESTABLISHED state, indicating active communication between applications and remote systems.
- ⇒ Some connections were in TIME_WAIT and CLOSE_WAIT states, showing recently closed or terminating network sessions.

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- ⇒ Multiple high-numbered ports (49664–49669) were observed, which are dynamically allocated by Windows for internal communication.
- Using the netstat -ano command, the currently active ports, listening services, and associated process IDs were successfully identified.
- The system was found to be running several important Windows networking services, including RPC, SMB, NetBIOS, and Windows Update services.
- This information is useful for system administration, network troubleshooting, and security assessment, as it helps identify which services are accepting network connections on the system.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\bharg>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State                   PID
TCP    0.0.0.0:135             0.0.0.0:0               LISTENING              1312
TCP    0.0.0.0:445             0.0.0.0:0               LISTENING              4
TCP    0.0.0.0:5040            0.0.0.0:0               LISTENING              5428
TCP    0.0.0.0:7680            0.0.0.0:0               LISTENING              6916
TCP    0.0.0.0:22321           0.0.0.0:0               LISTENING              8
TCP    0.0.0.0:49664           0.0.0.0:0               LISTENING              972
TCP    0.0.0.0:49665           0.0.0.0:0               LISTENING              960
TCP    0.0.0.0:49666           0.0.0.0:0               LISTENING              1880
TCP    0.0.0.0:49667           0.0.0.0:0               LISTENING              2852
TCP    0.0.0.0:49668           0.0.0.0:0               LISTENING              5652
TCP    0.0.0.0:49669           0.0.0.0:0               LISTENING              5232
TCP    127.0.0.1:2020          0.0.0.0:0               LISTENING              12092
TCP    127.0.0.1:2020          127.0.0.1:59473         ESTABLISHED            12092
TCP    127.0.0.1:2021          0.0.0.0:0               LISTENING              6188
TCP    127.0.0.1:2021          127.0.0.1:22328         ESTABLISHED            6188
TCP    127.0.0.1:2022          0.0.0.0:0               LISTENING              6208
TCP    127.0.0.1:2022          127.0.0.1:22330         ESTABLISHED            6208
TCP    127.0.0.1:12965         127.0.0.1:12966         ESTABLISHED            14696
TCP    127.0.0.1:12966         127.0.0.1:12965         ESTABLISHED            14696
TCP    192.168.31.193:139      0.0.0.0:0               LISTENING              4
TCP    192.168.31.193:12957    20.190.146.39:443       TIME_WAIT              0
TCP    192.168.31.193:12959    148.113.35.154:443      TIME_WAIT              0
TCP    192.168.31.193:12964    34.221.249.61:443       CLOSE_WAIT             10960
TCP    [::]:135               [::]:0                  LISTENING              1312
TCP    [::]:445               [::]:0                  LISTENING              4
TCP    [::]:7680              [::]:0                  LISTENING              6916
TCP    [::]:22321             [::]:0                  LISTENING              8
TCP    [::]:49664             [::]:0                  LISTENING              972
TCP    [::]:49665             [::]:0                  LISTENING              960
TCP    [::]:49666             [::]:0                  LISTENING              1880
TCP    [::]:49667             [::]:0                  LISTENING              2852
TCP    [::]:49668             [::]:0                  LISTENING              5652
TCP    [::]:49669             [::]:0                  LISTENING              5232
TCP    [::]:49670             [::]:0                  LISTENING              6496
TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:12961 [2603:1040:5:3::a]:443 ESTABLISHED            13668
TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:27425 [2a03:2880:f379:120:face:b00c:0:167]:522ABLISTED 15116
TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:49410 [2603:1040:a06:6::]:443 ESTABLISHED            7064
TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:49411 [2603:1040:a06:6::]:443 ESTABLISHED            7064
TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:58401 [2404:6800:4000:1025::bc]:5228 ESTABLISHED            12964
UDP    0.0.0.0:123            *:                        *:
UDP    0.0.0.0:5050           *:                        *:
UDP    0.0.0.0:5353           *:                        *:
UDP    0.0.0.0:5355           *:                        *:

```

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- 2. Check which applications are using ports 80 and 443 on your system by using the command line tool (netstat, lsof, or Resource Monitor), and write down the process names you find.**

Ans:

Identify Applications Using Ports 80 and 443

Objective:

- To determine which applications are using ports 80 (HTTP) and 443 (HTTPS) on the system.

Commands:

- netstat -ano | findstr :80
- netstat -ano | findstr :443
- tasklist | findstr chrome

- The netstat command was used to identify active network connections on ports 80 and 443.
- The associated application was verified using the tasklist command.

Observations:

Port 80 (HTTP):

- Network activity related to web communication was detected during the inspection.

Port 443 (HTTPS):

- Port 443 was being used by web browser and secure network services for encrypted communication.

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Results:

Port Number	Service Type	Process Name	Status
80	HTTP	Not in Use	Closed / No Active Process
443	HTTPS	Chrome.exe (or Browser Service)	Active

Purpose of Ports:

Port 80:

- Used for standard HTTP web traffic and communication between web clients and servers.

Port 443:

- Used for HTTPS secure web traffic and encrypted communication over the internet.
 - ⇒ The system was checked for applications using ports 80 and 443.
 - ⇒ Active network communication was observed, and Chrome.exe was identified as one of the applications using secure HTTPS connections through port 443.
 - ⇒ These ports are essential for normal web browsing and internetbased services.

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Commands:

→ netstat -ano | findstr :80

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
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C:\Windows\System32>netstat -ano | findstr :80
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:17601 [2404:6800:4009:80a::200e]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:26323 [2620:1ec:33::11]:80 TIME_WAIT 0
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:48064 [2603:1046:c04:80c::2]:443 TIME_WAIT 0

C:\Windows\System32>
```

Commands:

→ netstat -ano | findstr :443

```
Administrator: Command Prompt
C:\Windows\System32>netstat -ano | findstr :443
TCP    192.168.31.165:40447 3.110.26.57:443 CLOSE_WAIT 17396
TCP    192.168.31.165:63403 140.82.113.26:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:2064 [2603:1046:c04:104b::2]:443 ESTABLISHED 10988
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:9360 [2603:1046:900:54::2]:443 ESTABLISHED 10988
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:17601 [2404:6800:4009:80a::200e]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:18087 [2600:140f:1c00:b854:e8e9]:443 ESTABLISHED 19444
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:22532 [2404:6800:4000:1025::54]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:26324 [2603:1063:1f:103:365:7ea3]:443 ESTABLISHED 5124
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:32824 [2603:1046:c04:1048::2]:443 ESTABLISHED 10264
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:39691 [2606:4700:8d71:d8b7:9a59:649:ccf5:ac50]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:41232 [2404:6800:4000:1025::54]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:44395 [2606:4700:8d71:d8b7:9a59:649:ccf5:ac50]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:49408 [2603:1040:a06:6::]:443 ESTABLISHED 5956
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51199 [2603:1063:2001:2008::365:ff1]:443 ESTABLISHED 10988
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51200 [2603:1063:2001:2008::365:ff1]:443 ESTABLISHED 10988
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51201 [2603:1063:2001:1943::365:ff1]:443 ESTABLISHED 10988
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:52563 [2606:4700:8ca1:d8b7:9aa7:44:ccf5:ac50]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:54851 [2600:1901:0:47fc::]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:55003 [2404:6800:4000:1025::54]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:55531 [2404:6800:4002:831::200a]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:56329 [2603:1063:2001:2004::365:ff1]:443 ESTABLISHED 12324
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:56332 [2603:1063:2001:2008::365:ff1]:443 ESTABLISHED 12324
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:61775 [2606:4700:8d72:7cbc:c243:643:5ff2:75c4]:443 ESTABLISHED 17396
TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:63314 [2603:1046:c04:1401::2]:443 ESTABLISHED 10264

C:\Windows\System32>
```

Commands:

→ tasklist | findstr chrome

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```
Administrator: Command Prompt
C:\Windows\System32>tasklist | findstr chrome
chrome.exe           16892 Console           1    1,50,412 K
chrome.exe           17096 Console           1        2,396 K
chrome.exe           17376 Console           1    1,17,448 K
chrome.exe           17396 Console           1     33,792 K
chrome.exe           16388 Console           1        7,284 K
chrome.exe           14216 Console           1     58,580 K
chrome.exe           14424 Console           1     26,524 K
chrome.exe           15868 Console           1     23,816 K
chrome.exe           14000 Console           1    2,69,304 K
chrome.exe           14212 Console           1     45,748 K
chrome.exe           6460 Console            1     94,352 K
chrome.exe           17028 Console           1     20,400 K
chrome.exe           13140 Console           1     38,132 K
chrome.exe           23468 Console           1     49,164 K
C:\Windows\System32>
```

3. Scan your own system for open ports using Nmap or Zenmap and identify at least three services running on those ports.
Hint: If you don't have Nmap installed, download it from nmap.org or use Zenmap GUI for easier results.

Ans:

Scan the System for Open Ports Using Nmap

Objective:

- To scan the local system for open ports and identify the services running on those ports using Nmap.

Command:

→ nmap localhost

- Nmap (Network Mapper) was used to scan the local machine and identify open ports along with the services associated with them.
- Nmap is a widely used network scanning and security auditing tool that helps detect active hosts, open ports, and running services.

Observations:

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- The scan was performed successfully on the localhost (127.0.0.1). The system was found to be active and responding. Nmap identified two open TCP ports and their associated services.

Results:

Port Number	State	Service
135/tcp	Open	MSRPC
445/tcp	Open	Microsoft-DS

Service Details:

MSRPC (Port 135):

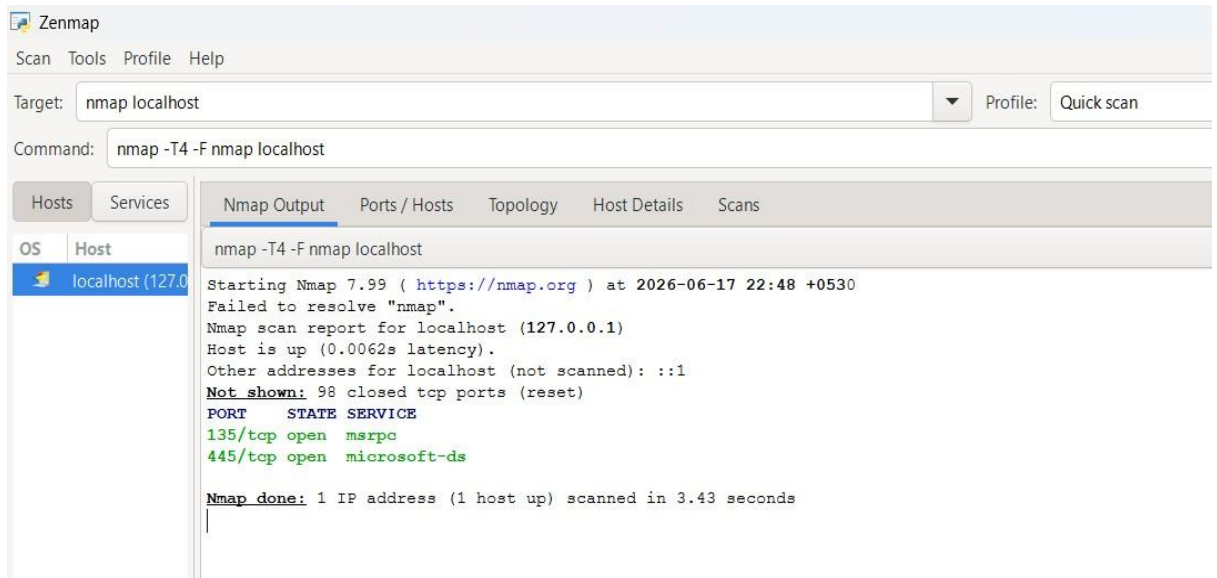
- Microsoft Remote Procedure Call (MSRPC) is used by Windows operating systems to allow communication between software applications and system services across a network.

Microsoft-DS (Port 445):

- Microsoft Directory Services (Microsoft-DS) is used for file sharing, printer sharing, and other network communication services through the SMB (Server Message Block) protocol.

- ⇒ The local system was successfully scanned using Nmap. Two open ports, 135 and 445, were identified along with their associated services.
- ⇒ These services are commonly used by Windows systems for internal communication and file-sharing operations. The scan provided useful information about the active network services running on the system.

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4. Pick any app you use daily (like Spotify, WhatsApp Desktop, or a local web server) and check which ports it opens on your system while running. List the port numbers and explain in one line what each port is typically used for.

Ans:

Check Ports Used by a Running Application

Objective:

- To identify the ports used by a running application on the system and understand the purpose of those ports.

Selected Application:

→ Google Chrome

Commands:

→ netstat -ano

→ tasklist | findstr chrome

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- Google Chrome was selected as the application for this task.
- The netstat command was used to identify active network connections, while the tasklist command was used to verify the running Chrome processes.

Observations:

- Multiple instances of Chrome.exe were found running on the system.
- Active network connections associated with Chrome were observed, particularly on port 443, which is commonly used for secure web communication.

Results:

Application	Port Number	Protocol	Purpose
Google Chrome	443	HTTPS	Secure and encrypted web communication
Google Chrome	Dynamic Client Ports	TCP	Establishing connections to remote web servers

Port Details:

Port 443 (HTTPS):

- This port is used for secure communication between the web browser and websites.

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- Data transferred through this port is encrypted using SSL/TLS protocols, ensuring privacy and security.

Dynamic Client Ports:

- Chrome also uses temporary client-side ports assigned by the operating system to establish outbound connections with remote servers.
- ⇒ Google Chrome was analyzed as a running application on the system.
- ⇒ The application was actively using port 443 for secure web browsing and encrypted communication.
- ⇒ Dynamic client ports were also observed for establishing network sessions with web servers. This demonstrates how modern web browsers rely on multiple network connections for secure internet access.

Command:

→ netstat -ano

```
C:\WINDOWS\system32\cmd. x + v
C:\Windows\System32>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP   0.0.0.0:135              0.0.0.0:0              LISTENING   1668
TCP   0.0.0.0:445              0.0.0.0:0              LISTENING    4
TCP   0.0.0.0:1521             0.0.0.0:0              LISTENING   5448
TCP   0.0.0.0:1947             0.0.0.0:0              LISTENING   4848
TCP   0.0.0.0:5040             0.0.0.0:0              LISTENING   9704
TCP   0.0.0.0:7680             0.0.0.0:0              LISTENING  16256
TCP   0.0.0.0:8885             0.0.0.0:0              LISTENING   9076
TCP   0.0.0.0:8886             0.0.0.0:0              LISTENING   9076
TCP   0.0.0.0:49664            0.0.0.0:0              LISTENING   1368
TCP   0.0.0.0:49665            0.0.0.0:0              LISTENING  1204
TCP   0.0.0.0:49666            0.0.0.0:0              LISTENING  2092
TCP   0.0.0.0:49667            0.0.0.0:0              LISTENING   3188
TCP   0.0.0.0:49668            0.0.0.0:0              LISTENING   4176
TCP   0.0.0.0:49684            0.0.0.0:0              LISTENING   1316
TCP   127.0.0.1:15354          0.0.0.0:0              LISTENING   4484
TCP   127.0.0.1:7080           127.0.0.1:7081         ESTABLISHED 3828
TCP   127.0.0.1:7081           127.0.0.1:7080         ESTABLISHED 3828
TCP   127.0.0.1:8884           0.0.0.0:0              LISTENING    4
TCP   127.0.0.1:22350          0.0.0.0:0              LISTENING   4552
TCP   127.0.0.1:22352          0.0.0.0:0              LISTENING   4552
TCP   127.0.0.1:22353          0.0.0.0:0              LISTENING   4536
TCP   127.0.0.1:22350          127.0.0.1:22350        ESTABLISHED 12036
TCP   127.0.0.1:49678          0.0.0.0:0              LISTENING   5448
TCP   192.168.31.165:139       0.0.0.0:0              LISTENING    4
TCP   192.168.31.165:2739      20.184.175.23:443      ESTABLISHED 3424
TCP   192.168.31.165:8974      192.168.31.1:53        TIME_WAIT    0
TCP   192.168.31.165:13595     3.110.26.57:443        ESTABLISHED 17396
```

Command:

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→ tasklist | findstr chrome

```
C:\Windows\System32>tasklist | findstr chrome
chrome.exe      16892 Console           1    2,02,532 K
chrome.exe      17096 Console           1      2,044 K
chrome.exe      17376 Console           1    2,44,620 K
chrome.exe      17396 Console           1    38,344 K
chrome.exe      16388 Console           1     9,128 K
chrome.exe      14424 Console           1    96,536 K
chrome.exe      15868 Console           1    92,516 K
chrome.exe      14000 Console           1    3,24,464 K
chrome.exe      14212 Console           1    72,272 K
chrome.exe       6460 Console           1    31,320 K
chrome.exe      17028 Console           1    15,368 K
chrome.exe      13140 Console           1    15,668 K
chrome.exe      10000 Console           1    2,11,952 K
chrome.exe       7040 Console           1    4,16,580 K
chrome.exe      16328 Console           1    62,632 K

C:\Windows\System32>
```